

PET 8mm-to-6mm Transformation Methods

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Summary

ADNI PET scans have been smoothed using scanner-specific filter functions to produce Step4 Preprocessed PET images of uniform isotropic resolution of 8mm FWHM, the approximate resolution of the lowest resolution scanners used in ADNI [1, 2]. With the recent improvement in the resolution of PET scanners of sites participating in ADNI4, all PET scans acquired since Sept 1 2022 have been preprocessed at a uniform resolution of 6mm instead of 8mm FWHM. All the older early scans have been re-processed at 6mm by the Michigan group, except for some PET scans acquired with older incompatible PET scanners with low resolution.

Image resolution has a small but quantifiable effect on PET SUVRs. To maximize data compatibility while taking advantage of the improvement in scanner resolution, our group at UC Berkeley performed spatial resolution conversions of the numerical data (imputation) for scans that could only be reconstructed at 8mm but not 6mm FWHM resolution.

We provide two separate numerical datasets for [18F] florbetapir (FBP), [18F] florbetaben (FBB), and [18F] flortaucipir (FTP):

- (i) 8mm resolution dataset for amyloid and tau PET scans acquired pre-Sept1 2022 only
- (ii) 6mm resolution dataset for the complete set of historical and prospective amyloid and tau PET scans (which include imputed SUVRs for 320 FBP, 19 FBB and 91 FTP scans that could not be processed at 6mm)

This methods document explains the 8mm-to-6mm transformation for scans acquired with older incompatible PET scanners with low resolution in the 6mm datasets. The rows with imputed SUVR values are labelled with "8mm" in the IMAGE_RESOLUTION column. We imputed SUVR for the primary outcomes only (as listed in Table 1) so the other numerical data are blank.

We only provide a single 8mm resolution dataset for [18F] fluorodeoxyglucose (FDG) because acquisition of FDG has halted in ADNI4. We do not impute partial volume corrected (PVC) FTP SUVRs because our PVC algorithm has been updated in addition to the 8mm-to-6mm switch, complicating direct comparison of the pre- and post-Sept 1 2022 FTP PVC datasets.



Methods

To derive 8mm SUVR vs 6mm SUVR relationships, we selected a subset of ADNI FBP, FBB and FTP baseline scans that could be pre-processed at both 8mm and 6mm FWHM resolution. The SUVRs were generated from Step4 Preprocessed images, smoothed to a uniform resolution of 8mm or 6mm FWHM, processed using our standard ADNI MRI-dependent pipeline [3, 4].

For each tracer, we generated 8mm-to-6mm transformation equations for the SUVRs of the tracer-specific primary ROIs and recommended longitudinal reference region (as listed in Table 1). These reversible equations are based on total least square regression of 8mm SUVR against 6mm SUVR shown in Figure 1, 2 and 3. Given the high correlations between SUVRs ($R^2 \geq 0.96$), we applied these equations to transform 8mm SUVRs to equivalent 6mm SUVRs.

For FBP and FBB, we also calculated Centiloids and amyloid +/- thresholds that were derived from our existing approaches (see UC Berkeley Amyloid PET Methods document) using the imputed 6mm SUVR.

Note that these equations were developed for the specific primary ROIs intensity normalized by the cross-sectional reference region (FBB & FBP: whole cerebellum; FTP: inferior cerebellum). The equations are not validated for use with other ROIs. The primary ROI SUVRs can be "re-intensity normalized" once by dividing them by their recommended longitudinal reference regions SUVRs, since this re-normalization cancels out the original intensity normalization.

Table 1. 8mm-to-6mm SUVR transformation equations for tracer-specific primary ROIs normalized to their respective cross-sectional reference regions. Slopes and intercepts are calculated based on total least square (TLS) regression of 6mm SUVR against 8mm SUVR (see Figure 1, 2, and 3). R^2 is a pseudo R-squared value derived by comparing TLS-predicted values to the actual values.

Tracer	SUVR Variable (norm. by cross-sectional ref)	Region Type	Slope	Intercept	R^2
FBP	SUMMARY_SUVR	Target	1.094	-0.113	0.996
	COMPOSITE_REF_SUVR	Longitudinal Ref	1.065	-0.070	0.982
FBB	SUMMARY_SUVR	Target	1.097	-0.116	0.998
	COMPOSITE_REF_SUVR	Longitudinal Ref	1.045	-0.041	0.968
FTP	CTX_ENTORHINAL_SUVR	Target	1.116	-0.121	0.987
	META_TEMPORAL_SUVR	Target	1.083	-0.092	0.998
	ERODED_SUBCORTICALWM_SUVR	Longitudinal Ref	0.987	-0.003	0.972



Figure 1. Linear regressions of FBP 6mm against 8mm whole cerebellum-normalized cortical summary and composite reference SUVR.

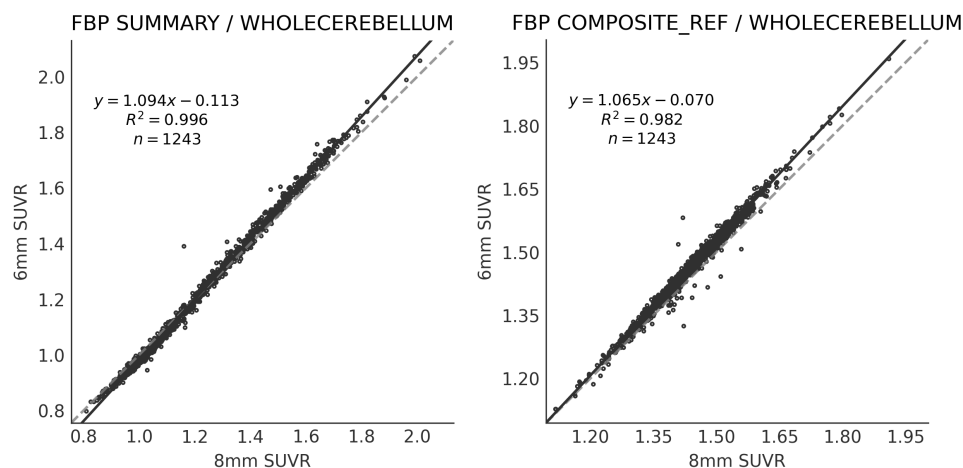


Figure 2. Linear regressions of FBB 6mm against 8mm whole cerebellum-normalized cortical summary and composite reference SUVR.

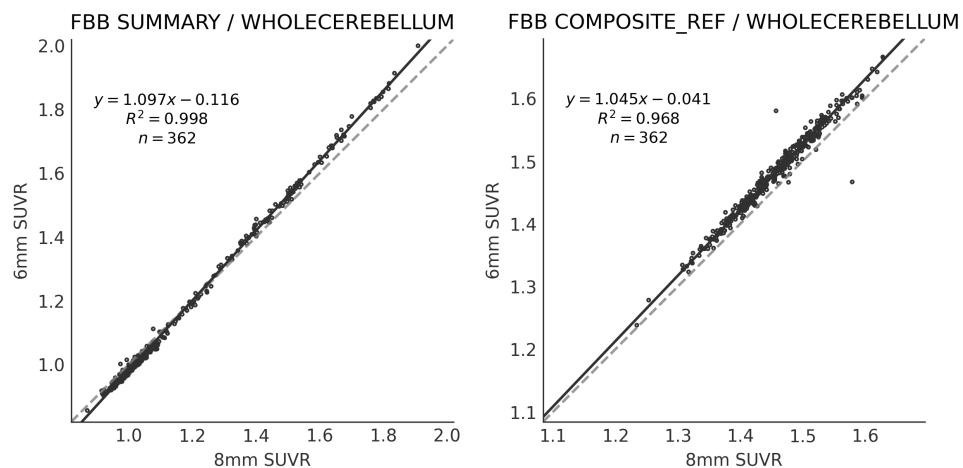
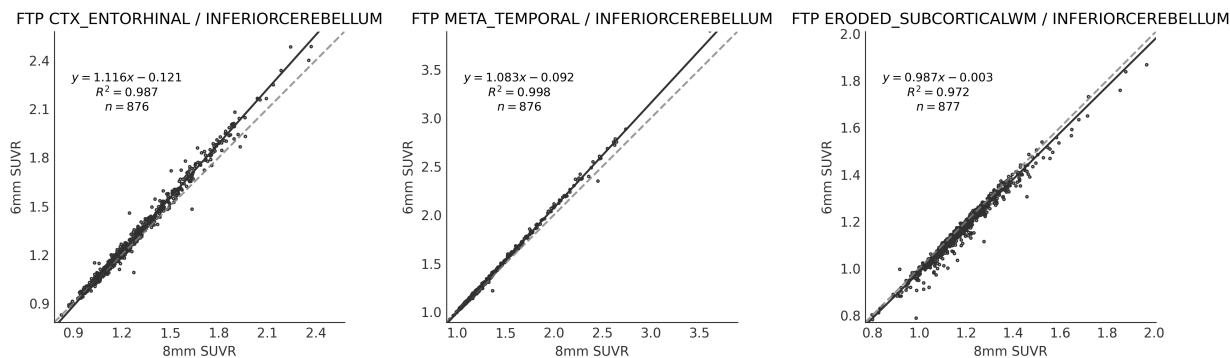


Figure 3. Linear regressions of FTP 6mm against 8mm inferior cerebellum-normalized entorhinal, temporal meta-ROI and eroded subcortical WM SUVR.



Version Information

This is our first PET 8mm-to-6mm Transformation standalone methods document.

Dataset Information

This methods document applies to the following datasets available from the ADNI repository:

TBLNAME	Dataset Name	Date Submitted
UCBERKELEY_AMY_6MM	UC Berkeley - Amyloid PET 6mm Res Analysis	25 May 2023 (*1st version)
UCBERKELEY_TAU_6MM	UC Berkeley – Tau PET 6mm Res Analysis	25 May 2023 (*1st version)

*New data will be uploaded to the 6mm datasets periodically.

Users can refer to its “update_stamp” column to check when the dataset is last updated. The “PROCESSDATE” column represents the date when the quantification pipeline was run. If a PET image has been re-processed (e.g., when a new MRI becomes available), its numerical data will be updated along with new PROCESSDATE and update_stamp.

Data dictionaries can be found at Download Study Data > Study Info > Data Dictionary [ADNI1,GO,2,3]. Use TBLNAME column to filter the corresponding data dictionaries from the long spreadsheet.

References

1. ADNI: PET Pre-processing. ADNI. adni.loni.usc.edu/methods/pet-analysis-method/pet-analysis/#pet-pre-processing-container
2. Joshi A, Koeppe RA, Fessler JA. Reducing between scanner differences in multi-center PET studies. *Neuroimage*. 2009 May 15;46(1):154-9. doi: 10.1016/j.neuroimage.2009.01.057
3. Lee JQ, Murphy A, Ward TJ, Harrison TM, Landau SM, Jagust WJ. Amyloid PET processing methods. LONI ADNI. 2023.
4. Landau SM, Lee JQ, Ward TJ, Murphy A, Baker SL, Harrison TM, Jagust WJ. Tau PET processing methods. LONI ADNI. 2023

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